

CHRIST (Deemed to be University), Bengaluru

Department of Computer Science

Project Synopsis

Course: CSC481-5 — Computer Science Project

Semester VI

ScholarLens

Offline SLM-Based Research Literature Review Agent

Sl. No.	Student Name	Register No.
1.	Aditya Mehta	2440204
2.	Jitvan Chadha	2440222
3.	Ritesh K R	2440233

Guide: Dr. New Begin

Class: 5BSC CS

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Synopsis

Title: ScholarLens — An Offline SLM-Based Research Literature Review Agent

Problem Statement: Conducting a thorough literature review is a critical yet time-intensive task for researchers and students. The process involves reading, organising, and synthesising large numbers of published papers to understand the state of the art in a given field. While several AI-powered tools (e.g. Elicit, Semantic Scholar) have emerged to assist with this process, they universally depend on cloud infrastructure and internet connectivity. This raises concerns around data privacy, accessibility in low-connectivity environments, and recurring costs that are prohibitive for students. There is currently no integrated, offline tool that combines document ingestion, intelligent search, automated summarisation, and research gap identification in a single privacy-preserving application.

Proposed Solution: ScholarLens is a fully offline, privacy-first research literature review agent powered by locally-running Small Language Models (SLMs). The system allows users to upload research papers in PDF format, which are then parsed, chunked, and embedded into a local vector database. Users can search their paper collection using a hybrid approach that combines semantic similarity with keyword-based retrieval. The system further provides SLM-driven capabilities including paper summarisation, key claim extraction, thematic clustering, and research gap analysis—all executed entirely on the user's machine without any data leaving the local environment.

Technology Stack: The backend is developed in Python using the FastAPI framework. The frontend is a React.js single-page application built with Vite. Language model inference is handled by Ollama running locally, while ChromaDB serves as the persistent vector database. Sentence-Transformers provide the embedding generation, and BM25 is used for keyword-based retrieval. PDF processing utilises PyMuPDF and pdfplumber.

Modules:

1. **Document Ingestion** — PDF parsing, text chunking, and metadata extraction.
2. **Knowledge Base** — Embedding generation and vector storage via ChromaDB.
3. **Search & Retrieval** — Hybrid semantic + keyword search with ranked results.
4. **Analysis** — SLM-powered summarisation, claim extraction, and theme clustering.
5. **Gap Finder** — Research gap identification and direction suggestion.

Expected Outcome: A functional, self-contained web application that enables researchers and students to manage, search, analyse, and derive insights from their research paper collection entirely offline, ensuring complete data privacy and zero recurring costs.